# Topic 3: DC Circuits and RC Transients

## Formula sheet

$$I = dQ/dt \text{ , } J = I/A = nqv_d$$

$$V = IR \text{ , } R = \blacksquare L/A \text{ , } \blacksquare(T) = \blacksquare_0[1 + \blacksquare(T - T_0)]$$

$$P = VI = I^2R = V^2/R$$

$$\text{Series: } R_{eq} = \blacksquare R_i \text{ Parallel: } 1/R_{eq} = \blacksquare(1/R_i)$$

$$C = Q/V \text{ , } C = \blacksquare\blacksquare_0 A/d \text{ , } U = 1/2CV^2 = Q^2/2C$$

$$\text{Series C: } 1/C_{eq} = \blacksquare(1/C_i) \text{ Parallel C: } C_{eq} = \blacksquare C_i$$

$$V_{term} = E - Ir \text{ , } I = E/(R + r)$$

$$\text{Charging: } q(t) = CE[1 - e^{-t/RC}] \text{ , } i(t) = (E/R)e^{-t/RC}$$

$$\text{Discharging: } q(t) = Q_0e^{-t/RC} \text{ , } V_C(t) = V_0e^{-t/RC}$$

$$\blacksquare = RC \text{ , } t_{1/2} = RC \ln 2 \blacksquare 0.693\blacksquare$$

## Worked example 1 - drift velocity

A copper wire of diameter 1.02 mm carries 1.67 A. Free electron density n = 8.49 × $10^{28}$ $m^{-3}$.

$$A = \blacksquare(d/2)^2 = \blacksquare(0.51 \times 10^{-3})^2 = 8.171 \times 10^{-7} \ m^2$$

$$v_d = I/(nqA) = 1.67 / (8.49\times10^{28} \times 1.602\times10^{-19} \times 8.171\times10^{-7})$$

$$v_d = 1.50 \times 10^{-4} \ m/s$$

Roughly 0.15 mm/s, about twenty minutes to cross one metre. The lamp lights instantly because the field is established at nearly the speed of light and starts every electron in the circuit moving at once.

## Worked example 2 - combination network

Resistors of 6.0 ■ and 3.0 ■ in parallel, in series with 4.0 ■, across 12.0 V of negligible internal resistance.

$$R_p = (6.0)(3.0)/(6.0 + 3.0) = 2.0 \blacksquare \text{ , } R_{eq} = 2.0 + 4.0 = 6.0 \blacksquare$$

$$I = V/R_{eq} = 12.0/6.0 = 2.0 \ A$$

$$V_p = IR_p = 4.0 \ V \blacksquare I_6 = 0.667 \ A \text{ , } I_3 = 1.333 \ A$$

$$P_4 = 16.0 \ W \text{ , } P_6 = 2.67 \ W \text{ , } P_3 = 5.33 \ W \text{ , } \blacksquare P = 24.0 \ W = VI \text{ [ok]}$$

## Worked example 3 - Kirchhoff

Two loops sharing a branch: $E_1$ = 12.0 V with $R_1$ = 2.00 ■, $E_2$ = 6.00 V with $R_2$ = 3.00 ■, shared $R_3$ = 4.00 ■.

$$I_1 + I_2 = I_3$$

$$12.0 - 2.00I_1 - 4.00I_3 = 0$$

$$6.00 - 3.00I_2 - 4.00I_3 = 0$$

Eliminating $I_3$:

$$6.00I_1 + 4.00I_2 = 12.0$$

$$4.00I_1 + 7.00I_2 = 6.00$$

$$\blacksquare\ I_1 = 2.31\ A\ ,\ I_2 = -0.462\ A\ ,\ I_3 = 1.85\ A$$

The negative result for $I_2$ means the assumed direction was wrong and the 6 V cell is being charged. The diagram is not corrected; the sign carries the information.

## Worked example 4 - RC charging

R = 100 k■, C = 10.0 ■F, E = 12.0 V, initially uncharged.

$$\blacksquare = RC = 1.00\ s\ ,\ Q_f = CE = 120\ \blacksquare C$$

$$q(0.500) = 120[1 - e^{-0.5}] = 120(0.39347) = 47.2\ \blacksquare C$$

$$i(0) = E/R = 120\ \blacksquare A\ ,\ i(2.00) = 120e^{-2} = 16.2\ \blacksquare A$$

$$0.90 = 1 - e^{-t/\blacksquare}\ \blacksquare\ t = -\blacksquare\ ln(0.10) = 2.30\ s$$

## Worked example 5 - energy in charging

Independent of R, exactly half the energy supplied by the source during charging is dissipated in the resistance:

$$W_{battery} = QE = CE^2\ ,\ U_{cap} = 1/2CE^2\ ,\ W_R = 1/2CE^2$$

## Derivation - q(t) from the loop rule

$$E - iR - q/C = 0\ ,\ i = dq/dt$$

$$R(dq/dt) = E - q/C = (CE - q)/C$$

$$dq/(CE - q) = dt/(RC)$$

$$-ln(CE - q) = t/RC + const$$

$$q = 0\ at\ t = 0\ \blacksquare\ const = -ln(CE)$$

$$ln[(CE - q)/CE] = -t/RC$$

$$q(t) = CE(1 - e^{-t/RC})$$

This derivation has appeared in three of the last five papers. Reproduce every line.

## Problems

- A 15.0 ■ and a 25.0 ■ resistor in parallel, in series with 10.0 ■, across 24.0 V with r = 0.800 ■. Find all currents and voltages, and verify the power balance.
- Aluminium, ■ = $2.82 \times 10^{-8}$ ■×m. Find the diameter of a 20.0 m wire of resistance 0.500 ■.
- A 4.70 ■F capacitor at 50.0 V is connected in parallel with an uncharged 10.0 ■F capacitor. Find the common final voltage and the energy lost. Account for the loss.
- The current in an RC circuit falls to 25.0% of its initial value in 3.40 ms. With C = 2.20 ■F, find R.
- Show that maximum power reaches a load when R = r, and state the efficiency at that point.